

Session Objectives

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At the conclusion of the session, you will be able to:

- Describe and explain the most important features defining cardiac valvular disease
- Compare and contrast the causes and consequences of **aortic valve stenosis**
- Define **mitral valve prolapse and mitral annular calcification**
- Demonstrate an understanding of the etiopathogenesis of **acute rheumatic fever and rheumatic heart disease**
- Summarize the pathology of **acute and subacute infective endocarditis**, as well as their associated macroscopic, microscopic, and clinical consequences
- Differentiate between **nonbacterial endocarditis and Libman-Sacks endocarditis**

Valvular disease

Valvular disease may result in stenosis, or both.

- **Stenosis:** it is caused by a **primary cuspal abnormality** determined by a chronic process: calcification or valve scarring.
- **Insufficiency:** is caused by an **intrinsic disease of the valve cusps** (e.g., endocarditis) or **disruption of the supporting structures** (tendinous cords, papillary muscles).

It can appear abruptly, as with chordal rupture, or insidiously as a consequence of leaflet scarring and retraction

Valvular disease can involve only one valve (the mitral valve being the most common target), or more than one valve.

Turbulent flow through diseased valves → murmurs

Severe stenosis → thrills (may be palpated)

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The Outcome: depends on:

- the valve involved
- the degree of impairment
- the tempo of its development
- the compensatory mechanisms.

Rapid destruction of an aortic valve cusp by infection can cause massive regurgitation and the abrupt onset of cardiac failure.

Rheumatic mitral stenosis usually progresses over years, and its clinical effects are well tolerated until late in the course.

Calcific Aortic Stenosis

Most cases

- asymptomatic
- calcifications on a CXR

In some patients, it can be severe enough to require surgical intervention.

The incidence parallels longevity.

In normal valves: when patients reach their 70s and 80s;

In bicuspid valves: when patients reach their 40s and 50s

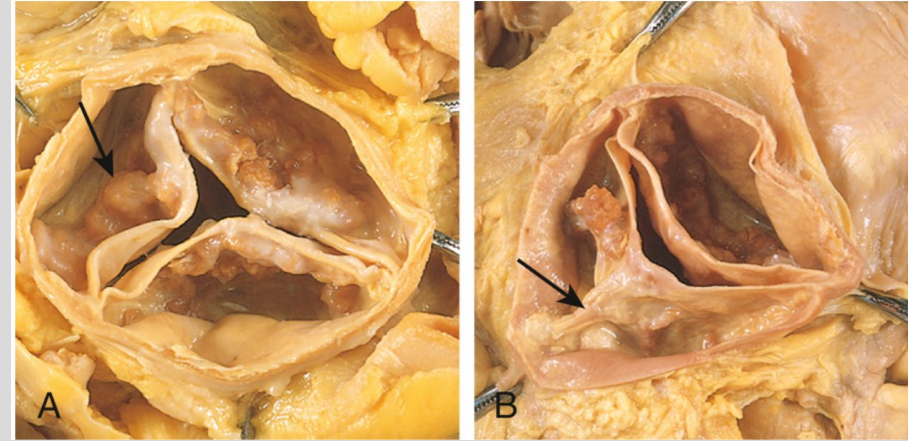
Pathogenesis

Simple progressive age-associated “wear and tear”

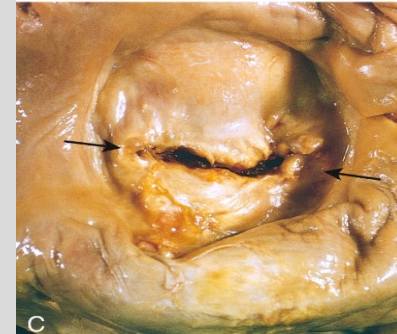
Cuspal fibrosis and calcification

Equivalent of atherosclerosis

It is associated with hyperlipidemia, hypertension, inflammation, and other factors implicated in atherosclerosis

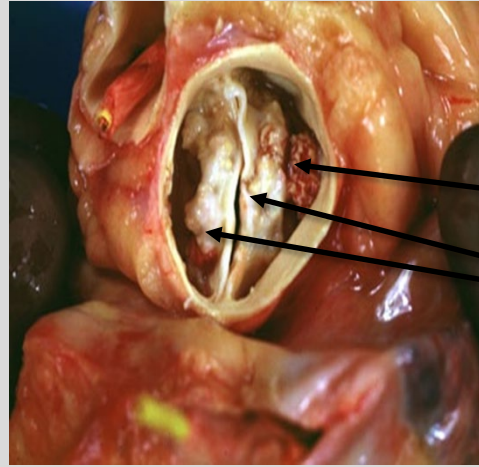
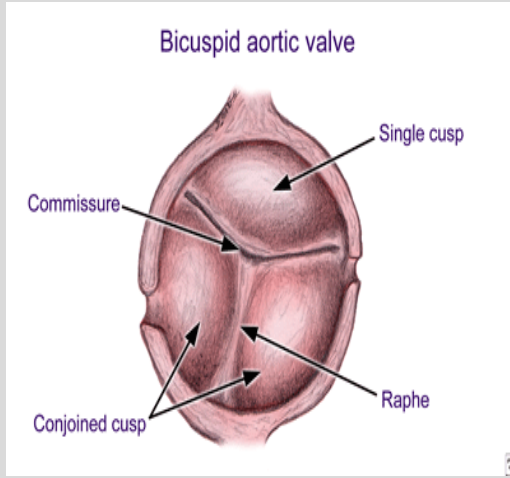


- A) Calcific aortic stenosis of a previously normal. Nodular masses of calcium are heaped up within the sinuses of Valsalva (arrow) .
- B) Calcific aortic stenosis occurring on a congenitally bicuspid valve. One cusp has a partial fusion at its center, called a raphe (arrow)



- C) Fused commissures in rheumatic aortic valve stenosis

Congenital Bicuspid Aortic Valve



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thrombus

calcifications

- In ~1% of the population;
- ~50% of cases of aortic stenosis in adults
- Familial clustering → associated with mutation in NOTCH1
- Morphology: two functional cusps, usually unequal in size
 - Larger valve → has midline raphe (from incomplete commissural separation during development)
 - Raphe = major site where calcific deposits occur

Mitral Valve Prolapse (MVP)

One or both mitral leaflets are “floppy” and prolapse—they balloon back into the left atrium during systole.

Associated conditions

Marfan syndrome
Ehlers-Danlos syndrome
polycystic kidney disease
fragile X syndrome

Etiology

Myxomatous degeneration
-Primary spontaneous
-Secondary to a connective tissue disorder
Rheumatic fever
Myocardial infarction

Pathogenesis

myxomatous degeneration of the mitral valve

- valvular leaflets become abnormal secondary to
 - excessive degradation of collagen and elastin by proteolytic enzymes
 - accumulation of proteoglycans
- the chordae tendinae become weakened and elongated
- annular dilation of the mitral valve

connective tissue disorders

- deficiency in collagen and elastin cause leaflets to be mostly thin and translucent

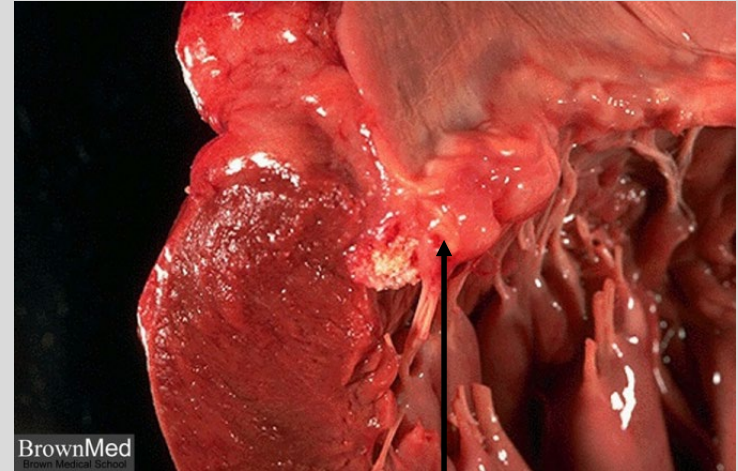
Mitral Annular Calcification (MAC)

- MAC is most commonly an incidental finding requiring no specific treatment other than assessment and modification of cardiovascular risk factors.
- See in women > 60 years old; especially in those with myxomatous mitral valve or elevated left ventricle pressure
- **Pathogenesis:** degenerative calcific deposits that develop in the fibrous annulus.
- MAC may be associated with atherosclerosis.

The most important intervention for most patients with MAC is evaluation and treatment of standard cardiovascular risk factors.

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Irregular, stony hard, occasionally ulcerated nodules at the base of the leaflets

ARF Clinical Features

3% of patients: typically appears 10 days to 6 weeks after a group A streptococcal infection.

It occurs most often in children between 5 and 15 years of age

Major Criteria for Dx

J - Joint Pain

♥ - Heart

N - Nodules

E - Erythema marginatum

S - Sydenham chorea

Acute carditis:

Pericardial friction rubs

Tachycardia

Arrhythmias.

Myocarditis

Mitral valve insufficiency

Heart failure.

1% of affected individuals die of fulminant RF involvement of the heart.

Arthritis typically begins with migratory polyarthritis (accompanied by fever) in which one large joint after another becomes painful and swollen for a period of days and then subsides spontaneously, leaving no residual disability

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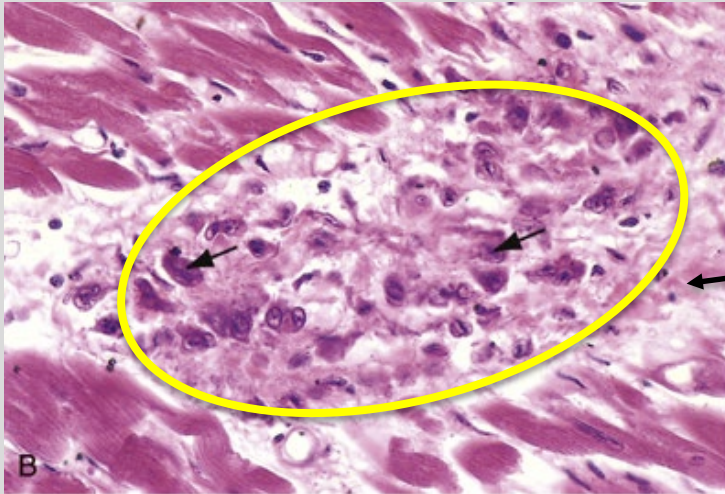
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ARF Morphology

Myocarditis

Distinctive lesions in the heart—called **Aschoff bodies**—are composed of foci of T lymphocytes, occasional plasma cells, and plump activated macrophages called **Anitschkow cells**.

Diffuse inflammation and Aschoff bodies may be found in any of the three layers of the heart, resulting in pericarditis, myocarditis, or endocarditis (pancarditis).



Aschoff body in a patient with acute carditis.

The myocardium exhibits a circumscribed nodule of mixed mononuclear inflammatory cells with associated necrosis. Within the inflammation, large activated macrophages show prominent nucleoli, as well as chromatin condensed into long, wavy ribbons (caterpillar cells) represented by the arrows.

These are also known as Anitschkow cells.

RHD Clinical Features

It is diagnosed years or even decades after the initial episode of RF

Cardiac murmurs

Cardiac hypertrophy and dilation

Heart failure

Arrhythmias

Thromboembolic complications

Endocarditis

Variable prognosis

Surgical repair of diseased valves

Replacement of diseased valves

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Chronic Rheumatic Heart Disease Morphology

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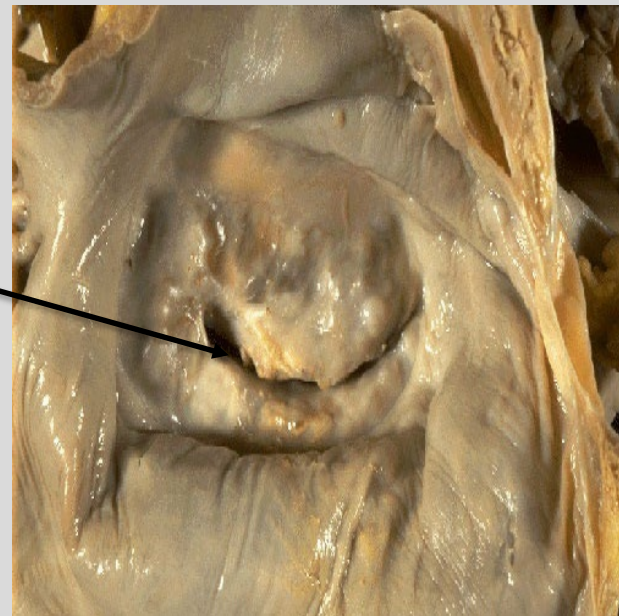
The mitral valve is virtually always involved in chronic RHD;
It is affected in isolation in roughly two-thirds of cases.
It is affected along with the aortic valve in another 25%.

Tricuspid valve and pulmonary valve are only rarely affected

In **rheumatic mitral stenosis**, calcification and fibrous bridging across the valvular commissures create “fish mouth” stenoses.

If severe, the left atrium progressively dilates and may harbor **mural thrombi** that can **embolize**.

Congestive changes in the lungs may induce pulmonary vascular and parenchymal changes; over time, these can lead to **right ventricular hypertrophy**.



Infective Endocarditis (IE)

=microbial infection of the heart valves or the mural endocardium that leads to the formation of vegetations composed of thrombotic debris and organisms, often associated with destruction of the underlying cardiac tissues.

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Acute and subacute forms

Acute IE : **infection of a previously normal heart valve** by a highly virulent organism (e.g., *Staphylococcus aureus*) that rapidly produces destructive lesions.

- Difficult to cure with antibiotics alone and often require surgery
- Substantial morbidity and even mortality.

Subacute IE: **infection of a deformed valve** by organisms with lower virulence (e.g., viridans streptococci) that cause insidious infections;

- Overall less destruction.
- Protracted course of weeks to months
- Cures can often be achieved with antibiotics alone

Infective Endocarditis Risk Factors

- Rheumatic Heart Disease
- Mitral Valve Prolapse
- Degenerative Calcific Valvular Stenosis
- Prosthetic Valves
- Congenital Defects
- Immunocompromised patients (i.e., drug abuse disorder, neutropenia, immunodeficiency, malignancy, diabetes)



Nonbacterial Thrombotic Endocarditis

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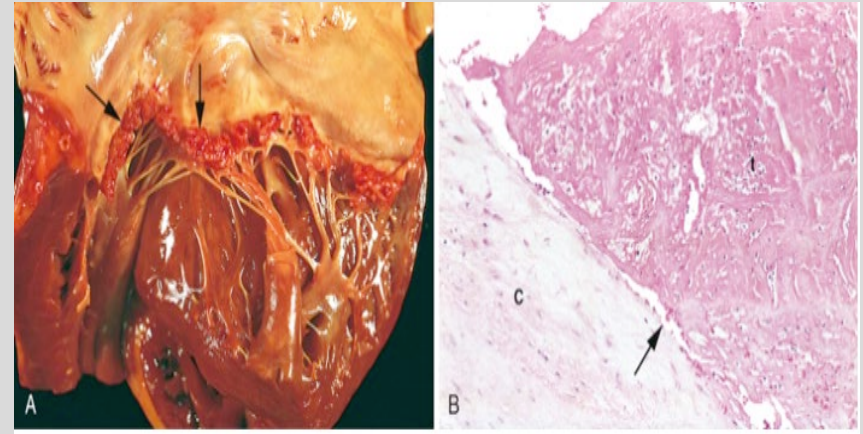
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Valvular damage is not a prerequisite for NBTE; this condition usually occurs on previously **normal valves**.

NBTE is often seen in those with **cancer-particularly mucinous adenocarcinomas, sepsis or associated with hypercoagulable states**

- Disseminated intravascular coagulation
- Hyperestrogenic states

Endocardial trauma, as from an indwelling catheter, is a well-recognized predisposing condition.



A. Nearly complete row of thrombotic vegetations along the line of closure of the mitral valve leaflets (arrows)

B. Histology of bland vegetation, with virtually no inflammation in the valve cusp (c) or the thrombotic deposit (t). The thrombotic material (**fibrin, platelets, blood components**) is only loosely attached to the cusp (arrow)

Summary

General features defining cardiac valvular disease

Aortic Stenosis

- Congenital bicuspid valve
- Acquired
- Equivalent of the age-related atherosclerosis
- "wear and tear" mechanism

Mitral Valve Prolapse

- floppy leaflets, myxomatous degeneration

Mitral Annular Calcification

- most of the time no therapy, associated with coronary and atherosclerotic disease.

Acute Rheumatic Fever vs Rheumatic Heart Disease

- group A beta hemolytic streptococci
- Aschoff bodies, Anitschikow cells
- ARF JONES major criteria
- RHD-the only cause of acquired mitral stenosis "fish-mouth" pattern (please also see table on slide 4)

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Infective Endocarditis

- acute vs subacute
- etiology, morphology, clinico-pathological consequences.
- Streptococcus viridans vs Staphylococcus aureus
- Janeway lesions, Roth spots, Nail bed or Splinter hemorrhages, Osler nodules

Nonbacterial thrombotic endocarditis

- mucinous adenocarcinoma, hypercoagulable states, sterile vegetations/thrombi, infarcts

Libman-Sacks endocarditis

- SLE, immune complex depositions, valvulitis, fibrinoid necrosis.